

## Task for Backend Developer

# Introduction

The task is to be solved by using two languages (for example: node.js and golang), but one can switch the golang with the other language, in case this will speed up the solving of the task!

Regarding the selection of the programming languages:

- 1) one is expected to be node.js unless explicitly agreed
- 2) The second language of choice is expected to be Golang unless explicitly agreed

*Note: Please inform the interviewer about your language selection or agree with the interviewer before starting the task.*

This task requires the applicant to implement a service that can receive, process and store the information about an mp4 file. This service is to be composed of two components

- Rest based API service that will handle the user interaction
- Processing service that will handle the parsing of the file itself

These services should communicate with each other using NATS, and should store the information in an SQL database.

# The API Service

The API service should provide the following API calls

- List all files provided to the service
- List details of a single file
- Start processing
- Delete file information

List all files should return a list of files that were provided to the API service with the status of the processing (Processing, Failed, Successful)

Single File Details should list the file and the location of the processed data if successful, otherwise it should provide information about why the processing failed

Start Processing API call should send the path to the mp4 file to the API, which needs to be sent to the processing service. This should be an absolute path to the file on the disk, and should be accessible to both services.

Delete File information should delete the file information and the processing data.

Additionally, API service should be able to handle NATS messages incoming from the Processing Service in order to properly store the information about the processing.

## The Processing Service

Service needs to process mp4 file and extract from its initialization segment of the MP4 file. Initialization segment needs to be stored to a new file and its path should be returned over NATS to the API service.

This is a very brief explanation of the structure of MP4 files, and the parsing algorithm:

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MP4 files are made up of boxes, like these:

[box1\_size][box1\_type][box1\_bytes][box2\_size][box2\_type][box2\_bytes]...

Box size is 32-bit, unsigned, big-endian integer, giving you the size of the box in bytes, including the bytes for the size, the type, and the data.

The type is also 32-bit, or 4 bytes, telling you the type of the box, as ASCII text.

Example:

[36][ftyp][28 bytes of ftyp data][787][moov][779 bytes of moov data]

The total size of the "ftyp" box is 36 bytes. The data portion is 28 bytes, which is 36 minus 4 bytes for the box size and 4 bytes for the box type.

The total size of the "moov" box is 787 bytes. The data portion is 779 bytes, which is 787 minus 4 bytes for the box size and 4 bytes for the box type.

We expect to find precisely the "ftyp" box followed by the "moov" box. These two represent the initialization segment. We consider everything afterwards to be the media segment. We don't inspect the structure of the MP4 file in any greater detail.

## Final Notes

- Please provide documentation describing the services and their structure, in addition to documentation describing how to run the exercise
  - If using docker compose, please provide clear instructions on expected volume mounts
- If any additional features not mentioned in this document are implemented, please explain the addition and rationale for the same
- Document any unresolved issues with the project

## Links

- Sample MP4 file - [http://devel.uniqcast.com/general/mp4\\_testing/video.mp4](http://devel.uniqcast.com/general/mp4_testing/video.mp4)